The stationary-start maintenance model and its exact delay-period predictions

Working note

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Abstract

One change to the maintenance model — initialising the Ornstein–Uhlenbeck maintenance noise at its stationary law rather than at zero — makes the transient irrelevant to the delay-period analysis window, and every prediction the paper tests then follows from the *stationary* Kac–Rice rate by a short corollary. The note is organised accordingly: Sections 1–4 establish that the transient can be neglected, and state precisely under what conditions; Sections 5–7 work with the stationary rate alone and derive the predictions. Every statement has been verified numerically (Section 9).

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1 The model

Everything before stimulus offset is unchanged. Maintenance begins at $\tau = t_{\text{offset}} + L_{\text{motion}}$, at which time the gate-facing signal y equals the maintained target m. During maintenance,

$$\mathrm{d}\varepsilon(t) = -\kappa\,\varepsilon(t)\,\mathrm{d}t + \sigma_\varepsilon\,\mathrm{d}W_\varepsilon(t), \tag{1}$$

$$\mathrm{d}y(t) = \lambda\,[\,m + \varepsilon(t) - y(t)\,]\,\mathrm{d}t, \tag{2}$$

and $H := y - m$ obeys $\mathrm{d}H = \lambda(\varepsilon - H)\,\mathrm{d}t$.

Initial conditions. $H(\tau) = 0$ is *forced*: the filter output is continuous and m is defined as the value of y at τ. It is not a modelling choice. By contrast $\varepsilon(\tau)$ is free; the present model sets it to 0. We propose instead

$$\varepsilon(\tau) \sim \mathrm{N}\big(0,\ \sigma_\varepsilon^2/2\kappa\big) \quad \text{independent of } W_\varepsilon \tag{ss}$$

— the stationary law of (1). Call (ss) the *stationary start* and $\varepsilon(\tau) = 0$ the *cold start*.

Why. ε represents fluctuations intrinsic to holding a value. Nothing makes such a process sit exactly at its mean at the instant maintenance begins, and no measurement constrains it. As Corollary 6 shows, the cold start is the sole source of a slow $(1/\kappa)$ transient in a system whose only other timescale is the fast filter $1/\lambda$; under (ss) the model's delay-period predictions become exact rather than asymptotic.

Remark 1. (ss) removes the mechanism that gives Model 1 its only delay-period ramp. That is not a way of disarming Model 1: Section 4 bounds what *any* residual non-stationarity can buy it, and the bound is far below the data.

Notation. $\rho := t - \tau$ is elapsed maintenance time; β/δ is the speed of temporal preparation, with $\beta = 0$ under Model 1; $d(t) = g - |m| - (\beta/\delta)(t - t_{\text{offset}})^+$ is the distance to the gate; φ, Φ are the standard normal density and c.d.f., and $g(u) := u\Phi(u) + \varphi(u)$, so $g' = \Phi$ and $g'' = \varphi$. Write $s^2 := \sigma_\varepsilon^2/2\kappa$ and

$$A_\infty := \frac{\lambda s^2}{\lambda + \kappa}, \qquad B_\infty := \frac{\lambda^2 \sigma_\varepsilon^2}{2(\lambda + \kappa)}, \qquad \omega := \sqrt{B_\infty/A_\infty} = \sqrt{\lambda\kappa}, \qquad a_\infty := \frac{d}{\sqrt{A_\infty}}. \tag{3}$$

2 Exact second-order moments

Proposition 2 (exact moments)**.** *Under* (1)–(ss) *with* $H(\tau) = 0$*, put*

$$X(\rho) := \frac{(\lambda + \kappa)\big(e^{-2\lambda\rho} - e^{-(\lambda+\kappa)\rho}\big)}{\lambda - \kappa} \qquad \big(= -2\lambda\rho\, e^{-2\lambda\rho} \quad \textit{when } \lambda = \kappa\big). \tag{4}$$

Then for every $\rho > 0$*,*

$$\boxed{\frac{\sigma_H^2}{A_\infty} = 1 + X - e^{-(\lambda+\kappa)\rho}, \qquad \frac{c_H}{A_\infty} = -\lambda X, \qquad \frac{\sigma_{\dot H}^2}{B_\infty} = 1 + \frac{\lambda}{\kappa}\Big(X + e^{-(\lambda+\kappa)\rho}\Big).} \tag{5}$$

In particular $\sigma_H^2(0) = 0$*,* $c_H(0) = 0$*,* $\sigma_{\dot H}^2(0) = \frac{\lambda+\kappa}{\kappa} B_\infty$*, and all three converge to* (3) *with* $c_H(\infty) = 0$*.*

Proof. (ε, H) is a two-dimensional linear diffusion with $F = \left(\begin{smallmatrix} -\kappa & 0 \\ \lambda & -\lambda \end{smallmatrix}\right)$ and $Q = \mathrm{diag}(\sigma_\varepsilon^2, 0)$, so its covariance solves $\dot P = FP + PF^\top + Q$ with $P(0) = \mathrm{diag}(s^2, 0)$. The system is triangular:

$$\dot P_{\varepsilon\varepsilon} = -2\kappa P_{\varepsilon\varepsilon} + \sigma_\varepsilon^2, \qquad P_{\varepsilon\varepsilon}(0) = s^2, \tag{6}$$

$$\dot P_{\varepsilon H} = -(\kappa + \lambda) P_{\varepsilon H} + \lambda P_{\varepsilon\varepsilon}, \qquad P_{\varepsilon H}(0) = 0, \tag{7}$$

$$\dot P_{HH} = 2\lambda(P_{\varepsilon H} - P_{HH}), \qquad P_{HH}(0) = 0. \tag{8}$$

By (6), $\dot P_{\varepsilon\varepsilon}(0) = -2\kappa s^2 + \sigma_\varepsilon^2 = 0$, so $P_{\varepsilon\varepsilon} \equiv s^2$: *the stationary start freezes the first component exactly.* Then (7) gives $P_{\varepsilon H} = A_\infty\big(1 - e^{-(\lambda+\kappa)\rho}\big)$, and (8), integrated from 0,

$$P_{HH} = 2\lambda \int_0^\rho e^{-2\lambda(\rho - v)} P_{\varepsilon H}(v)\mathrm{d}v = A_\infty \left[1 - e^{-2\lambda\rho} - 2\lambda \frac{e^{-(\lambda+\kappa)\rho} - e^{-2\lambda\rho}}{\lambda - \kappa}\right],$$

and writing $-2\lambda = -(\lambda+\kappa) - (\lambda-\kappa)$ in the numerator gives $P_{HH}/A_\infty - 1 = X - e^{-(\lambda+\kappa)\rho}$. Since $\dot{H} = \lambda(\varepsilon - H)$, $c_H = \lambda(P_{\varepsilon H} - P_{HH}) = -\lambda A_\infty X$ and $\sigma_{\dot{H}}^2 = \lambda^2(P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}) = B_\infty + \lambda^2 A_\infty\big(X + e^{-(\lambda+\kappa)\rho}\big)$, using $\lambda^2(s^2 - A_\infty) = B_\infty$ and $\lambda^2 A_\infty/B_\infty = \lambda/\kappa$. The $\lambda = \kappa$ forms follow by l'Hôpital. □

Remark 3. σ_H rises from 0; $\sigma_{\dot{H}}$ *falls* from $\frac{\lambda+\kappa}{\kappa}B_\infty$, because $\dot{H}(\tau) = \lambda\varepsilon(\tau)$ has the noise at full amplitude while H is still at zero. The position variance fills in while the velocity variance settles down.

3 How fast the transient dies

Lemma 4. *For $A, B > 0$ and $\rho > 0$, $\left|e^{-A\rho} - e^{-B\rho}\right| \le |A - B|\,\rho\, e^{-\min(A,B)\rho}$.*

Proof. The mean value theorem applied to $x \mapsto e^{-x\rho}$ on the interval with endpoints A, B gives $|A - B|\rho e^{-\xi\rho}$ for some ξ between them, and $e^{-\xi\rho} \le e^{-\min(A,B)\rho}$. □

Proposition 5 (transient bounds)**.** *Set $\mu := \lambda + \min(\lambda, \kappa)$, $E_1 := e^{-(\lambda+\kappa)\rho}$, $E_2 := e^{-2\lambda\rho}$, and*

$$\mathcal{D}(\rho) := \min\left\{\rho\, e^{-\mu\rho},\ \frac{E_1 + E_2}{|\lambda - \kappa|}\right\} \qquad (\textit{first entry only, when } \lambda = \kappa). \tag{9}$$

Then, for all $\lambda, \kappa, \rho > 0$,

$$\bar{\Delta} := (\lambda+\kappa)\mathcal{D} + E_1, \qquad \bar{\gamma} := \lambda(\lambda+\kappa)\mathcal{D}, \qquad \bar{e} := \min\left\{\frac{\lambda}{\kappa}\bar{\Delta},\ \frac{\lambda\big[(\lambda+\kappa)E_2 + 2\kappa E_1\big]}{\kappa|\lambda - \kappa|}\right\} \tag{10}$$

bound the three deviations:

$$\left|\frac{\sigma_H^2}{A_\infty} - 1\right| \le \bar{\Delta}, \qquad 0 \le \frac{c_H}{A_\infty} \le \bar{\gamma}, \qquad \left|\frac{\sigma_{\dot{H}}^2}{B_\infty} - 1\right| \le \bar{e}. \tag{11}$$

All three are decreasing on $\rho \ge 1$ when $\lambda \ge 1$.

Proof. Everything reduces to bounding $|E_1 - E_2|/|\lambda - \kappa|$, since by Proposition 2 $X = (\lambda + \kappa)(E_2 - E_1)/(\lambda - \kappa)$, $\sigma_H^2/A_\infty - 1 = X - E_1$, $c_H/A_\infty = -\lambda X$, and $\sigma_{\dot{H}}^2/B_\infty - 1 = (\lambda/\kappa)\big[(\lambda + \kappa)E_2 - 2\kappa E_1\big]/(\lambda - \kappa)$, this last by adding and subtracting $(\lambda+\kappa)E_1$ in the numerator. Two bounds are available and we take the smaller. *(a)* By Lemma 4 with $A = 2\lambda$, $B = \lambda + \kappa$ — so $|A - B| = |\lambda - \kappa|$ and $\min(A, B) = \mu$ — $|E_1 - E_2|/|\lambda - \kappa| \le \rho e^{-\mu\rho}$. *(b)* Trivially $|E_1 - E_2| \le E_1 + E_2$. Bound (a) is the sharp one when $\lambda \approx \kappa$, where (b) is singular; (b) is the sharp one when the two rates are far apart, where (a)'s worst-case exponent μ is pessimistic. The sign $X \le 0$ follows by inspection of (4), giving $c_H \ge 0$. For the third bound, the first entry of $\bar{e}$ comes from the X form and the second from bounding the two exponentials separately. Monotonicity: for the first entry, $\frac{\mathrm{d}}{\mathrm{d}\rho}(1 + c\rho)e^{-\mu\rho} = e^{-\mu\rho}(c - \mu - \mu c\rho)$ with $c = \lambda + \kappa \ge \mu$, negative for $\rho > (c - \mu)/(\mu c)$, and $(c - \mu)/(\mu c) \le 1/(2\lambda) \le 1/2$ when $\lambda \ge 1$; the second entry is a sum of decaying exponentials. □

Corollary 6 (relaxation is at least as fast as λ)**.** From Propositions 2 and 5. *All three moments converge to (3) at rate $\mu \ge \lambda$, uniformly in κ. The operator $P \mapsto FP + PF^\top$ has eigenvalues $\{-2\kappa, -(\lambda+\kappa), -2\lambda\}$, so a general initial condition relaxes as a combination of $e^{-2\kappa\rho}$ (the ε–ε block), $e^{-(\lambda+\kappa)\rho}$ (the cross term) and $e^{-2\lambda\rho}$ (the H–H block). Under (SS) the first has zero coefficient, because $P_{\varepsilon\varepsilon}$ is constant; the slower of the remaining two is $e^{-\mu\rho}$, which equals $e^{-2\lambda\rho}$ only when $\kappa \ge \lambda$.*

Remark 7 (the κ floor). The first two bounds in (11) are uniform in κ; the third carries λ/κ and is not. What it needs is weak: at $\rho = 1$ the deviation is $(\lambda/\kappa)e^{-2\lambda}$ to leading order, so relative accuracy ϵ needs $\kappa \gtrsim \lambda e^{-2\lambda}/\epsilon$, which at $\lambda = 10$, $\epsilon = 10^{-2}$ is $\kappa \geq 2 \times 10^{-6}$. Read through the correlation time: $\mathrm{Cov}(\varepsilon(t), \varepsilon(t+r)) \propto e^{-\kappa|r|}$ at stationarity, so $1/\kappa$ is the lag at which the maintenance noise decorrelates to $1/e$. A process with $1/\kappa$ of minutes is constant over the 3 s delay and over dozens of consecutive trials — a session-level drift, for which the model has no term. A floor $\kappa \geq 0.05\,\mathrm{s}^{-1}$ ($1/\kappa \leq 20$ s, about seven delay periods) is therefore a statement about what the term means, not a range imported from simulation.

4 Neglecting the transient

Let ν_{stat} denote the rate computed from the stationary moments (3), i.e. with $\sigma_H^2 = A_\infty$, $\sigma_{\dot H}^2 = B_\infty$, $c_H = 0$. This section bounds the error committed by using ν_{stat} in place of ν.

Theorem 8 (the cost of using the stationary rate)**.** *Fix $\rho \geq \rho_0 \geq 1$, evaluate $\bar\Delta, \bar\gamma, \bar e$ of (10) at ρ_0, and write $u_s := \beta/(\delta\sqrt{B_\infty})$, $c_g := \Phi(0)/g(0) = 1.2533$, and*

$$\eta := \bar e + \frac{\bar\gamma^2}{\omega^2(1-\bar\Delta)}. \tag{12}$$

Assume $\lambda \geq 1$, $\bar\Delta < 1$ and $\eta < 1$. Then, for both models,

$$\boxed{\left|\ln \nu(\rho) - \ln \nu_{\mathrm{stat}}(\rho)\right| \;\leq\; \mathcal{E} := \frac{(a_\infty^2+1)\bar\Delta}{2(1-\bar\Delta)} + \frac{\eta}{2(1-\eta)} + c_g\left[\frac{\bar\gamma\, a_\infty}{\omega(1-\bar\Delta)\sqrt{1-\eta}} + u_s\left(\frac{1}{\sqrt{1-\eta}} - 1\right)\right].} \tag{13}$$

Proof. Write $\Delta := \sigma_H^2/A_\infty - 1$, $\gamma := c_H/A_\infty$, $E := \sigma_{\dot H}^2/B_\infty - 1$, so that by Proposition 5 and its monotonicity clause $|\Delta| \leq \bar\Delta$, $0 \leq \gamma \leq \bar\gamma$, $|E| \leq \bar e$ for all $\rho \geq \rho_0$. From the Kac–Rice formula (15) below,

$$\ln \nu - \ln \nu_{\mathrm{stat}} = -\tfrac{1}{2}\big(a^2 - a_\infty^2\big) - \tfrac{1}{2}\ln\frac{\sigma_H^2}{A_\infty} + \ln\frac{G}{G_s},$$

with $G = \sqrt{S}\,g(u)$, $G_s = \sqrt{B_\infty}\,g(u_s)$. We bound the three terms. Throughout, $|\ln(1+x)| \leq |x|/(1-|x|)$ for $|x| < 1$.

(1) $a^2 = d^2/\sigma_H^2 = a_\infty^2/(1+\Delta)$, so $\big|a^2 - a_\infty^2\big| = a_\infty^2|\Delta|/(1+\Delta) \leq a_\infty^2\bar\Delta/(1-\bar\Delta)$, and the first term is at most $a_\infty^2\bar\Delta/\big(2(1-\bar\Delta)\big)$.

(2) $\frac{1}{2}\big|\ln(1+\Delta)\big| \leq \bar\Delta/\big(2(1-\bar\Delta)\big)$. Adding to (1) gives the first bracket of (13).

(3a) $S = \sigma_{\dot H}^2 - c_H^2/\sigma_H^2 = B_\infty\big[(1+E) - \gamma^2/(\omega^2(1+\Delta))\big]$, using $A_\infty/B_\infty = \omega^{-2}$. Hence $|S/B_\infty - 1| \leq \eta$ with η of (12), so $S \geq B_\infty(1-\eta) > 0$ and $\frac{1}{2}\big|\ln(S/B_\infty)\big| \leq \eta/\big(2(1-\eta)\big)$.

(3b) $M = \beta/\delta + \gamma d/(1+\Delta) \geq \beta/\delta \geq 0$, so $u = M/\sqrt{S} \geq 0$; also $u_s \geq 0$. $(\ln g)' = \Phi/g$ and $(\ln g)'' = (\varphi g - \Phi^2)/g^2 = -D/g^2 < 0$, where $D := \Phi^2 - \varphi g$ satisfies $D' = \varphi(\Phi + ug) > 0$ and $D(-\infty) = 0$, so $D > 0$: g is log-concave and Φ/g is decreasing. Hence on $[0,\infty)$, $\sup(\ln g)' = \Phi(0)/g(0) = c_g$, and for $u, u_s \geq 0$

$$\big|\ln g(u) - \ln g(u_s)\big| \leq c_g\,|u - u_s|.$$

Split $u - u_s = \dfrac{M - \beta/\delta}{\sqrt{S}} + \dfrac{\beta}{\delta}\Big(\dfrac{1}{\sqrt{S}} - \dfrac{1}{\sqrt{B_\infty}}\Big)$. For the first, $M - \beta/\delta = \gamma d/(1+\Delta) \leq \bar\gamma a_\infty\sqrt{A_\infty}/(1-\bar\Delta)$ and $\sqrt{S} \geq \sqrt{B_\infty(1-\eta)}$, giving $\bar\gamma a_\infty/\big(\omega(1-\bar\Delta)\sqrt{1-\eta}\big)$. For the second, it equals $u_s\big|(1+x)^{-1/2} - 1\big|$ with $x = S/B_\infty - 1$, $|x| \leq \eta$, hence at most $u_s\big(1/\sqrt{1-\eta} - 1\big)$. Summing gives (13). □

Corollary 9 (what it costs on the measured quantities)**.** From Theorem 8. *Let f be the log rate on the* 20 *bin centres of the window* $[1, 3]$ *s, and* c_1, c_2 *the slope and quadratic coefficient of the least-squares quadratic fitted to it. These are linear functionals* $c_j = \sum_i w_i^{(j)} f_i$ *with* $\sum_i |w_i^{(\mathrm{slope})}| = 1.504$ *and* $\sum_i |w_i^{(2)}| = 2.916$*. Hence, if* $|f - f_{\mathrm{stat}}| \le \mathcal{E}$ *pointwise,*

$$\left|c_1 - c_1^{\mathrm{stat}}\right| \le 1.504\,\mathcal{E}\ \mathrm{s}^{-1}, \qquad \left|c_2 - c_2^{\mathrm{stat}}\right| \le 2.916\,\mathcal{E}\ \mathrm{s}^{-2}. \tag{14}$$

Remark 10 (why this step is needed)*.* Proposition 5 bounds the *moments*; the observables are a slope and a curvature, that is, *derivatives*, and a bound on a function does not bound its derivative. Theorem 8 bridges the gap by bounding $\ln \nu$ itself, pointwise, after which (14) is elementary because c_1 and c_2 are bounded linear functionals of the profile. Without this step, "the transient is negligible" is an assertion, not a theorem.

4.1 The conditions, and what justifies each

Theorem 8 needs exactly three things, and each can be argued without reference to the measured rate.

(C1) $\lambda \ge 5$–$6\,\mathrm{s}^{-1}$**, i.e.** $1/\lambda \le 170$–200 **ms.** $1/\lambda$ is the time constant of the low-pass filter carrying motor preparation to the gate. Dendauw et al. estimate $\lambda \approx 20\,\mathrm{s}^{-1}$ (50 ms) in an immediate-report task, so this is three to four times more permissive than the published value. It is in principle identifiable from these data, since λ sets how fast the rate follows the stimulus; that check has not been carried out.

(C2) $\kappa \ge 0.2\,\mathrm{s}^{-1}$**, i.e.** $1/\kappa \le 5$ **s.** $1/\kappa$ is the correlation time of the maintenance noise: $\mathrm{Cov}(\varepsilon(t), \varepsilon(t+r)) \propto e^{-\kappa|r|}$ at stationarity, so it is the lag at which the fluctuation has decorrelated to $1/e$. Requiring it to be at most one trial says that what is maintained fluctuates *within* a trial; a longer correlation time describes a drift carried across trials, for which the model has no term and which the randomised design averages out rather than expresses as a delay-period ramp. (C1) and (C2) are substitutes: $\lambda \to \lambda + 1$ buys about what $\kappa \to 2\kappa$ buys, so only one needs to be argued hard.

(C3) $a_\infty \le 5$**.** Not fitted. By (18) the expected number of partial activations per 2 s delay window is $2\nu_{\mathrm{stat}} = (\omega/\pi)e^{-a^2/2}$; at the largest ω in play this is one per 19 trials at $a = 3.3$, one per 246 at $a = 4$, and *one per* 22 000 *at* $a = 5$. So $a \le 5$ says only that the model must produce partial activations at all — the experiment comprises of order 10^4 trials — not that it must reproduce the observed rate.

With $\rho_0 = 1$ s, $a_\infty \le 5$ and $\beta/\delta \le 0.3$:

λ	$1/\lambda$	$\kappa \ge 0.2$	$\kappa \ge 0.5$	$\kappa \ge 1$
4	250 ms	9.6×10^{-1}	6.4×10^{-1}	4.1×10^{-1}
5	200 ms	3.5×10^{-1}	2.3×10^{-1}	1.4×10^{-1}
6	167 ms	1.3×10^{-1}	8.4×10^{-2}	5.0×10^{-2}
8	125 ms	1.9×10^{-2}	1.2×10^{-2}	6.7×10^{-3}
10	100 ms	2.8×10^{-3}	1.6×10^{-3}	9.1×10^{-4}

Values of $\mathcal{E}$. By (14) the stationary formula then misstates a slope by at most $1.504\,\mathcal{E}$ and a curvature by at most $2.916\,\mathcal{E}$. At $\lambda \ge 6$, $\kappa \ge 0.2$: $\mathcal{E} = 0.133$, so the slope error is $\pm 0.20\,\mathrm{s}^{-1}$ against an observed $+0.867\,\mathrm{s}^{-1}$; at $\lambda \ge 5$ it is ± 0.53, and at $\lambda \ge 8$, ± 0.029. *From here on we work with* ν_{stat} *and carry* $1.504\,\mathcal{E}$ *as the error bar on every slope.*

5 The stationary rate

Let $Z := s_{\text{resp}}H - d$, so that partial activations are upcrossings of zero by Z and $\dot{Z} = \dot{H} + \beta/\delta$. The Kac–Rice formula for a differentiable Gaussian process gives

$$\nu(t) = p_{Z(t)}(0)\,\mathbb{E}\big[\dot{Z}\mathbf{1}_{\{\dot{Z}>0\}} \mid Z = 0\big] = \frac{\varphi(a)}{\sigma_H}\,G, \qquad G = M\Phi(u) + \sqrt{S}\varphi(u) = \sqrt{S}\,g(u), \quad (15)$$

with $a = d/\sigma_H$, $M = \beta/\delta + (c_H/\sigma_H^2)d$, $S = \sigma_{\dot{H}}^2 - c_H^2/\sigma_H^2$ and $u = M/\sqrt{S}$.

Proposition 11 (the stationary rate)**.** *Substituting the stationary moments* (3) *($c_H = 0$) into* (15)*,*

$$\nu_{\text{stat}} = \varphi(a)\left[\frac{\beta}{\delta\sigma_H}\Phi\!\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right) + \omega\,\varphi\!\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right)\right], \quad (16)$$

whose bracket does not depend on d. Consequently:

(i) ***Model 2*** *($\beta > 0$): $a(\rho) = a(0) - \beta\rho/(\delta\sigma_H)$ is affine, so exactly*

$$\frac{\mathrm{d}\ln\nu_{\text{stat}}}{\mathrm{d}\rho} = \frac{\beta}{\delta\sigma_H}\,a(\rho), \qquad \frac{\mathrm{d}^2\ln\nu_{\text{stat}}}{\mathrm{d}\rho^2} = -\left(\frac{\beta}{\delta\sigma_H}\right)^2, \quad (17)$$

a downward parabola, the same one in all six conditions.

(ii) ***Model 1*** *($\beta = 0$): $M = 0$, $u = 0$, a constant, and*

$$\nu_{\text{stat}} = \frac{\omega}{2\pi}\,e^{-a^2/2}, \qquad \frac{\mathrm{d}\ln\nu_{\text{stat}}}{\mathrm{d}\rho} \equiv 0, \qquad \frac{\mathrm{d}^2\ln\nu_{\text{stat}}}{\mathrm{d}\rho^2} \equiv 0. \quad (18)$$

Proof. Set $c_H = 0$ in (15); then $\ln\nu_{\text{stat}} = \text{const} - a^2/2$ and (17) follows from a affine. For (ii) set $\beta = 0$: $M = 0$, $g(0) = 1/\sqrt{2\pi}$, $G = \sqrt{B_\infty}/\sqrt{2\pi} = \omega\sigma_H/\sqrt{2\pi}$, so $\nu_{\text{stat}} = \varphi(a)\omega/\sqrt{2\pi} = \omega e^{-a^2/2}/2\pi$, constant in ρ. □

Proposition 12 (level orders by distance)**.** From (16). *Under both models $\nu_{\text{stat}} = \varphi(a) \times$ (a factor independent of d), so ν_{stat} is strictly decreasing in a, with no condition whatever. Under Model 2 the slope $\beta a/(\delta\sigma_H)$ of* (17) *is strictly* increasing *in a. Hence level and slope order oppositely across conditions.*

Remark 13 (before the window). The same monotonicity holds during the transient, though it then needs a condition: from (15), $\partial\ln\nu/\partial d = -a/\sigma_H + c_H\Phi(u)/(\sigma_H^2 G)$ is negative iff $dG > c_H\Phi(u)$, and $G > M\Phi(u)$ with $dM = (\beta/\delta)d + c_H a^2 \geq c_H$ whenever $a \geq 1$. Not needed for anything below.

6 The standardised distance is measured

Inverting (18) — Model 1's own stationary rate formula, so using it to bound Model 1 is legitimate —

$$a = \sqrt{2\ln\!\left(\frac{\omega}{2\pi\nu}\right)}. \quad (19)$$

(C3) uses only the weak, data-free reading of this identity — $a \leq 5$, on pain of the model producing fewer than one partial activation in twenty thousand trials. Two sharper readings are available if one is willing to use the measured rate, and neither is needed above. First, with the observed delay-period rates 0.030–0.135 s^{-1} and (λ, κ) anywhere in the sampled range, (19) gives $a \leq 3.3$, which tightens every ceiling in Section 4.1 by about a third. Second, (19) is the rate–ramp trade-off in its sharpest form: the rate *measures* the standardised distance, so the ramp Model 1 could produce and the rate it must produce are not free of one another.

7 Predictions

The delay-period observations are: **(L1)** the level rises with duration; **(L2)** the level rises with coherence; **(L3)** the coherence effect on the level is larger after the long stimulus; **(S1)** the log-rate slope is $+0.867\,\mathrm{s}^{-1}$; **(S2)** it is shallower after the long stimulus, $0.896 \to 0.723$; **(S3)** shallower at high coherence, $0.871 \to 0.726$. Throughout, the error bars are those of Corollary 9; at $\lambda \geq 6$, $\kappa \geq 0.2$, $a_\infty \leq 5$ they are $\pm 0.20\,\mathrm{s}^{-1}$ on a slope.

Corollary 14 (L1, L2: the level)**.** From Proposition 12, plus the saturation argument of the main text, which concerns the distribution of m and is untouched by (ss). *Duration and coherence each raise the fraction of trials whose maintained target has reached the cap θ, hence lower d in the sense of first-order stochastic dominance; since ν_{stat} is decreasing in d at every ρ, and averaging over trials preserves the ordering, both raise the level. Predicted by* both *models:* not discriminating.

Corollary 15 (L3: the interaction)**.** From Proposition 12 and the same saturation argument. *As the capped fraction tends to one the coherence levels converge, and duration raises that fraction at every coherence, so the coherence effect on the level must* shrink *with duration. Predicted by both models, and* contradicted *by the data. Not discriminating; a genuine disagreement.*

Corollary 16 (S1: the slope)**.** From Proposition 11 and Corollary 9. *Model 1 predicts a slope of exactly* 0*, hence a measured slope of at most* $1.504\,\mathcal{E}$*:* $0.20\,\mathrm{s}^{-1}$ *at* $\lambda \geq 6$*,* 0.029 *at* $\lambda \geq 8$ *(with* $\kappa \geq 0.2$*,* $a_\infty \leq 5$*). Model 2 predicts* $\beta a(\rho)/(\delta\sigma_H) > 0$*, its only source of a ramp.* Discriminating: the observed $+0.867\,\mathrm{s}^{-1}$ exceeds Model 1's ceiling by a factor of 4.3 at $\lambda \geq 6$ and 30 at $\lambda \geq 8$.

Corollary 17 (S2, S3: the slope across conditions)**.** From Propositions 11 and 12, Corollaries 9 and 14. *Model 1 predicts a slope of zero in* every *condition, hence a condition difference of at most* $2 \times 1.504\,\mathcal{E}$*:* $0.40\,\mathrm{s}^{-1}$ *at* $\lambda \geq 6$*,* 0.058 *at* $\lambda \geq 8$*. Model 2 predicts a difference of definite sign: the slope increases in a while the level decreases in a, so any manipulation that raises the level lowers the slope, and by Corollary 14 duration and coherence both raise the level.* The observed differences 0.173 and 0.145 carry the sign Model 2 requires; they exceed Model 1's ceiling at $\lambda \geq 8$ but not at $\lambda \geq 6$, so this leg is the one that pays for a low λ.

Remark 18 (on the curvature). Under the cold start the curvature of the log rate discriminated, through the decay that Model 1's variance filling forces. Under (ss) there is no decay to force: both models predict a near-zero curvature (Model 1 exactly 0, Model 2 $-\frac{1}{2}(\beta/\delta\sigma_H)^2$, which is ≥ -0.034 at the observed slope with $a \leq 3.3$), and the observed -0.0405 ± 0.0416 is consistent with both. The curvature therefore carries nothing here and is dropped from the list of tested observations; the discriminating role passes to S2 and S3.

8 Summary

At $\lambda \geq 6$, $\kappa \geq 0.2$, $a_\infty \leq 5$ (Section 4.1):

	Model 1	Model 2	verdict
L1, L2 level	rises	rises	shared
L3 interaction	shrinks	shrinks	shared; both fail
S1 slope	0 (≤ 0.20)	$\beta a/\delta\sigma_H > 0$	**discriminating**, 4.3×
S2, S3 by condition	0 (≤ 0.40)	sign forced	sign only at this λ

At $\lambda \geq 8$ the two ceilings become 0.029 and 0.058, and S2–S3 discriminate by magnitude as well (3.0× and 2.5×).

9 Numerical verification

Claim	Test	Result
Prop. 2	vs. numerical integration of $\dot{P} = FP + PF^\top + Q$	max rel. err. 8.9×10^{-12}
Prop. 2	the three identities in terms of X	agree to 1.5×10^{-13}
Prop. 5	$0 \le -X \le (\lambda+\kappa)\rho e^{-\mu\rho}$ (50-digit)	worst ratio 1.000000
Prop. 5	$\lvert\sigma_H^2/A_\infty - 1\rvert \le \bar{\Delta}$	worst ratio 1.000000
Prop. 5	$c_H/A_\infty \le \bar{\gamma}$	worst ratio 1.000000
Prop. 5	$\lvert\sigma_{\dot{H}}^2/B_\infty - 1\rvert \le (\lambda/\kappa)\bar{\Delta}$	worst ratio 0.997226
Prop. 5	tightened $\bar{\Delta}, \bar{\gamma}, \bar{e}$ (50-digit)	worst ratios 1.000000 (all three)
Thm 8	$\lvert\ln\nu - \ln\nu_{\text{stat}}\rvert \le \mathcal{E}$, 71 079 points	0 violations, worst ratio 0.612
Cor. 9	the two functional norms 1.504, 2.916	computed exactly

Grids: $\lambda \in [0.3, 60]$, $\kappa \in [0.01, 60]$, $\rho \in [0.05, 6]$, $\beta/\delta \in \{0, 0.05, 0.3, 1, 3\}$, $a_\infty \in \{0.5, 1, 2, 3.3, 6, 10\}$. The X bounds were verified in 50-digit arithmetic because the combination $P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}$ cancels catastrophically in double precision; points where $\mathcal{E}$ falls below double-precision resolution (10^{-10}) were excluded from the Theorem 8 check for the same reason.

10 Consequences for the paper

Adopting (ss) removes from the appendix the convolution kernel K, the integrals $\int K^2$ and $\int K'^2$, the auxiliary $W = K' - K^3/2\sigma_H^2$, the log-concavity lemma, the non-stationary derivative of $\ln\nu$, and the transient content of the present Corollaries A.2 and A.3. What remains is Sections 2–4 — five short results establishing that the transient may be neglected — followed by the stationary rate and the predictions as one-line corollaries.

The cost is that theory and simulation must agree, so the 1200-draw parameter sweep, the admissibility screen and every effect size would have to be recomputed under (ss).